



LOGS and EXPO selective

Part A: Revision of Indices

2 unit

TEACHER'S COPY

Q 1

Sydney
Boys

Simplify $\frac{15^{n+1} \times 25 \times 5^{3n-4}}{9^{n-1} \times 25^{n-2}}$, expressing with positive indices.

$$\begin{aligned} &= \frac{2 \cdot 5^{n+1} \times 5^2 \times 5^{3n-4}}{3^{2n-2} \times 5^{2n-2}} \\ &= \frac{2 \cdot 5^{n+1+2+3n-4}}{3^{2n-2} \times 5^{2n-2}} \\ &= \frac{2 \cdot 5^{4n-1}}{3^{2n-2} \times 5^{2n-2}} \\ &= \frac{2 \cdot 5^{4n-1-2n+2}}{3^{2n-2}} \\ &= \frac{2 \cdot 5^{2n-1}}{3^{2n-2}} \\ &= \frac{2 \cdot 5^{2n-1} \cdot 5}{3^{2n-2} \cdot 5} \\ &= \frac{2 \cdot 5^{2n}}{3^{2n-2} \cdot 5} \\ &= \frac{2 \cdot 5^{2n-1}}{3^{2n-2}} \end{aligned}$$

Q 2

Barker

An expression which is **not** equal to $2^{3x} \times 3^{3x}$ is:

(A) 6^{3x}

(B) $2^{3x} \times 9^{\frac{3x}{2}}$

(C) $\frac{2^{3x} \times 3^{3x-1}}{3}$

(D) $\frac{2^{3x}}{3^{-3x}}$

Q 3

Barker

Simplify fully $\frac{16a^3 - 54b^3}{4a^2 + 6ab + 9b^2}$.

$$\begin{aligned} &(b) \frac{16a^3 - 54b^3}{4a^2 + 6ab + 9b^2} \\ &= \frac{2(8a^3 - 27b^3)}{4a^2 + 6ab + 9b^2} \\ &= \frac{2(2a-3b)(4a^2 + 6ab + 9b^2)}{4a^2 + 6ab + 9b^2} \\ &= 2(2a-3b) \end{aligned}$$

Q 4

James
Ruse

Solve: $\frac{5^{3n+3}}{25^{n-3}} = 125$

$$\begin{aligned} \frac{5^{3n+3}}{25^{n-3}} &= 125 \\ \frac{5^{3n+3}}{5^{2n-6}} &= 5^3 \\ 5^{3n+3-2n+6} &= 5^3 \\ 5^{n+9} &= 5^3 \\ n+9 &= 3 \\ n &= -6 \end{aligned}$$

Q 5

Baulkham
Hills

Simplify $\frac{8^{x-2}}{2^{3-x}}$

$$\begin{aligned} \frac{8^{x-2}}{2^{3-x}} &= \frac{2^{3(x-2)}}{2^{3-x}} \rightarrow (1) \\ &= \frac{2^{3x-6}}{2^{3-x}} \\ &= 2^{3x-6-3+x} \\ &= 2^{4x-9} \quad (1) \end{aligned}$$

Q 6

CSSA

Solve for x: $5^{x+2} = \frac{1}{\sqrt[3]{125}}$

$$\begin{aligned} 5^{x+2} &= \frac{1}{5} = 5^{-1} \\ x+2 &= -1 \\ \therefore x &= -3 \end{aligned}$$





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Part A: Revision of Indices

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Q 7

Sydney
Girls

Simplify $\frac{x^2 y z^{-2} \times 2(x^2 y^{-1} z)^2}{x y z}$.

$$\begin{aligned} & \frac{x^2 y z^{-2} \times 2(x^2 y^{-1} z)^2}{x y z} \\ &= \frac{x^2 y z^{-2} \times 2(x^4 y^{-2} z^2)}{x y z} \\ &= \frac{x^2 y z^{-2} \times 2x^4 y^{-2} z^2}{x y z} \\ &= \frac{2x^6 y^{-1} z^0}{x y z} \\ &= \frac{2x^5}{y^2 z} \end{aligned}$$

Q 8

Sydney
Boys

Simplify $\frac{x^2 + 8x^{-1}}{1 + 2x^{-1}}$, expressing with positive indices.

$$\begin{aligned} f) \frac{x^2 + 8x^{-1}}{1 + 2x^{-1}} &= \frac{x^2 + \frac{8}{x}}{1 + \frac{2}{x}} \\ &= \frac{\frac{x^3 + 8}{x}}{\frac{x + 2}{x}} \\ &= \frac{x^3 + 8}{x + 2} \\ &= \frac{(x+2)(x^2 - 2x + 4)}{(x+2)} \\ &= x^2 - 2x + 4 \end{aligned}$$

Q 9

Sydney
Grammar

Simplify $\frac{15^{n+1} \times 25 \times 5^{3n-4}}{9^{n-1} \times 25^{n-2}}$, expressing with positive indices.

$$\begin{aligned} &= \frac{15^{n+1} \times 25 \times 5^{3n-4}}{9^{n-1} \times 25^{n-2}} \\ &= \frac{(3 \times 5)^{n+1} \times 5^2 \times 5^{3n-4}}{(3^2)^{n-1} \times (5^2)^{n-2}} \\ &= \frac{3^{n+1} \times 5^{n+1} \times 5^2 \times 5^{3n-4}}{3^{2n-2} \times 5^{2n-4}} \\ &= \frac{3^{n+1} \times 5^{n+1+2+3n-4}}{3^{2n-2} \times 5^{2n-4}} \\ &= \frac{3^{n+1} \times 5^{4n-1}}{3^{2n-2} \times 5^{2n-4}} \\ &= \frac{3^{n+1-2n+2} \times 5^{4n-1-2n+4}}{1} \\ &= \frac{3^{n+3} \times 5^{2n+3}}{1} \end{aligned}$$

Q 10

St George

Find the value of x when $\frac{3 \cdot 5^x - 1}{5^x + 2} = 2$.

$$\begin{aligned} i) \frac{3 \cdot 5^x - 1}{5^x + 2} &= 2 & j) 3^x &= \frac{1}{9} \\ 3 \cdot 5^x - 1 &= 2 \cdot 5^x + 4 & \frac{1}{3^x} &= \frac{1}{9} \\ 3 \cdot 5^x - 2 \cdot 5^x &= 5 & \frac{1}{3^2} &= \frac{1}{9} \\ 5^x (3 - 2) &= 5 & \therefore x &= -2 \\ 5^x &= 5 \\ \therefore x &= 1 \end{aligned}$$



Q 1

Find the value of $\log_4 7.5$ to 3 significant figures.

Santa Sabina

$$\begin{aligned}\log_4 7.5 &= \frac{\log_e 7.5}{\log_e 4} \\ &= 1.4534 \dots \\ &= 1.45\end{aligned}$$

Q 2

Given that $\log_x 5 = 0.627$ and $\log_x 2 = 0.27$ find the value of $\log_x 10$.

Santa Sabina

(A) 0.897

(B) 2.32

(C) 0.169

(D) 0.357

Q 3

Find the value of $\log_6 5.8$ to two significant figures.

Santa Sabina

(A) 2.4

(B) 2.3

(C) 1.0

(D) 0.98

Q 4

If $\log_a 2 = n$ and $\log_a 5 = m$, which expression gives $\log_a 50$?

St Catherines

(A) $m^2 + n$

(B) $m^2 n$

(C) $2m + n$

(D) $2mn$

Q 5

Find the value of $\log_5 200 - 3 \log_5 2$.

St George Girls

(A) 1.4

(B) 2.0

(C) 3.2

(D) 2.5

Q 6

What is the approximate value of $\log_5 37$?

St Ignatius

(A) 1.26

(B) 2.24

(C) 2.99

(D) 3.48

Q 7

$\log_3 15 + \log_3 18 - \log_3 10 =$

Sydney Boys

(A) 1

(B) 2

(C) 3

(D) 0



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PART B: LOGARITHMIC LAWS

2 unit

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Q 8

Simplify

Sydney Boys

$$\frac{\log(a^3b^2) - \log(ab^2)}{\log \sqrt{a}}$$

$$\begin{aligned} (c) \quad & \frac{\log a^3 + \log b^2 - (\log a + \log b^2)}{\log a^{\frac{1}{2}}} \\ & \frac{3\log a + 2\log b - \log a - 2\log b}{\frac{1}{2}\log a} \\ & \frac{2\log a}{\frac{1}{2}\log a} = 4 \quad \textcircled{D} \end{aligned}$$

Q 9

Evaluate $\log_{2\sqrt{3}} 144$.

Sydney Boys

- (A) 2 **(B) 4** (C) 6 (D) 10

Q 10

$\frac{\log_5 125}{\log_5 5}$ simplifies to

Sydney Boys

- (A) $\log_5 25$ (B) $\log_5 120$ (C) 25 **(D) 3**

Q 11

If $\log_a b = 0.5$ then $\log_a \sqrt{b} =$

Sydney Girls

- (A) 0.125 **(B) 0.250** (C) 0.707 (D) 1.000

Q 12

Which of the following statements is INCORRECT?

Sydney Grammar

- (A) $\log a^n = n \log a$
(B) $\log ab = \log a + \log b$
(C) $\log(a - b) = \frac{\log a}{\log b}$
(D) $\log e = 1$

Q 13

If $\log_a 7 = x$ and $\log_a 2 = y$

Enterprise

An expression for $\log_a 56$ in terms of x and y is

- (A) $3xy$ (C) $3x + y$
(B) $x + y$ **(D) $3y + x$**



Q 14

Enterprise

$$\log_{\sqrt{6}} 36 =$$

- (A) 1 (B) 4 (C) 6 (D) 36

Q 15

Trinity

Given $\log_a 3 = 0.6$ and $\log_a 2 = 0.4$, find the value of $\log_a 18$.

$$\begin{aligned}\log_a 18 &= \log_a (3^2 \times 2) \\ &= \log_a 3^2 + \log_a 2 \\ &= 2\log_a 3 + \log_a 2 \\ &= 2 \times 0.6 + 0.4 \\ &= 1.6\end{aligned}$$

Q 16

Sydney Tech

For a certain base, a , you are given that $\log_a 5 = x$ and $\log_a 2 = y$

Find an expression for $\log_a 12.5$

2

$$\begin{aligned}\log_a 12.5 &= \log_a \left(\frac{25}{2}\right) \\ &= 2\log_a 5 - \log_a 2 \\ &= 2x - y\end{aligned}$$

Q 17

Santa Sabina

If $a = \log_e 4$ and $b = \log_e 8$, what is the value of $e^a e^b$?

- (A) 12 (B) 32 (C) $\log_e 12$ (D) $\log_e 32$

Q 18

Sydney Boys

Let $a = e^x$. Which expression is equal to $\log_e (a^2)$?

- (A) e^{2x} (B) e^{x^2} (C) $2x$ (D) x^2

Q 19

Barker

Given $\log_a 3 = 0.6$ and $\log_a 2 = 0.4$, find $\log_a 18$.

- (A) 1.8 (B) 1.6 (C) 3.0 (D) 0.74

Q 20

Barker

The value of $\log_6 2000$ is closest to:

- (A) 1.3 (B) 4.2 (C) 5.8 (D) 9.4



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PART C: SOLVING EQUATIONS

2 unit

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Q 1

Show that $4^{3n-1} = 6$ can be expressed as

Santa Sabina

$$n = \frac{\ln 6}{3 \ln 4} + \frac{1}{3}$$

$$\begin{aligned} 4^{3n-1} &= 6 \\ \ln 4^{3n-1} &= \ln 6 \\ (3n-1) \ln 4 &= \ln 6 \\ 3n-1 &= \frac{\ln 6}{\ln 4} \\ 3n &= \frac{\ln 6}{\ln 4} + 1 \\ n &= \frac{\frac{\ln 6}{\ln 4} + 1}{3} \end{aligned}$$

Q 2

Which expression will help correctly solve for x the equation: $3^x = 12$

Santa Sabina

(A) $x = \log(12 - 3)$

(B) $x = \frac{\log 12}{\log 3}$

(C) $x = \log 12 \times \log 3$

(D) $x = \frac{\log 3}{\log 12}$

[3]

Q 3

(i) Given $2 \log_3 x - \log_3(x+4) = 2$, show that x satisfies $x^2 - 9x - 36 = 0$.

(ii) Hence, or otherwise, solve the equation:

Santa Sabina

$$2 \log_3 x - \log_3(x-2) = 2$$

$$\begin{aligned} \text{(b) i. } 2 \log_3 x - \log_3(x+4) &= 2 \\ \log_3(x^2) - \log_3(x+4) &= 2 \\ \log_3\left(\frac{x^2}{x+4}\right) &= 2 \\ 3^2 &= \frac{x^2}{x+4} \\ 9x + 36 &= x^2 \\ x^2 - 9x - 36 &= 0 \\ \text{as required.} \end{aligned}$$

$$\begin{aligned} \text{ii. } 2 \log_3 x - \log_3(x-2) &= 2 \\ \log_3 x^2 - \log_3(x-2) &= 2 \\ \log_3\left(\frac{x^2}{x-2}\right) &= 2 \\ \frac{x^2}{x-2} &= 3^2 \\ \frac{x^2}{x-2} &= 9 \\ x^2 &= 9x - 18 \\ x^2 - 9x + 18 &= 0 \\ (x-3)(x-6) &= 0 \\ x &= 3, x = 6. \end{aligned}$$

Q 4

Solve $2 \log_2 x - \log_2(2x+6) = 1$.

St George Girls

$$\begin{aligned} \text{(d) } 2 \log_2 x - \log_2(2x+6) &= 1 \\ \log_2\left(\frac{x^2}{2x+6}\right) &= \log_2 2 \quad \text{--- (1)} \\ \therefore \frac{x^2}{2x+6} &= 2 \\ x^2 &= 4x + 12 \\ x^2 - 4x - 12 &= 0 \quad \text{--- (1)} \\ (x+2)(x-6) &= 0 \\ x &= -2 \text{ OR } x = 6 \\ \text{Test } x = -2: 2 \log_2(-2) &\text{ is not defined} \\ \therefore x = -2 &\text{ is not a solution} \\ \text{Test } x = 6: 2 \log_2 6 - \log_2 18 &= \log_2 \frac{6^2}{18} = \log_2 2 \\ &= 1 \\ \therefore x = 6 &\text{ is the only solution. --- (1)} \end{aligned}$$



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PART C: SOLVING EQUATIONS

2 unit

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Q 5

What is the solution to the equation $\log_e(x+2) - \log_e x = \log_e 4$?

St George
Girls

- (A) $\frac{2}{5}$ (B) $\frac{2}{3}$ (C) $\frac{3}{2}$ (D) $\frac{5}{2}$

Q 6

Solve for x if $2\log_e x = \log_e 3 + \log_e(x+6)$.

Sydney
Boys

$$\begin{aligned} 2\ln x &= \ln 3 + \ln(x+6) \\ \ln x^2 &= \ln 3(x+6) \\ \therefore x^2 &= 3x + 18 \\ x^2 - 3x - 18 &= 0 \\ (x-6)(x+3) &= 0 \\ x &= 6, -3. \text{ (Clearly } x \neq -3) \\ \therefore \underline{x = 6} \end{aligned}$$

Q 7

Solve for x : $\log_5(2x+1) - \log_5 x = 2$

Sydney
Boys

$$\begin{aligned} \log_5(2x+1) - \log_5 x &= 2 \\ \log_5\left(\frac{2x+1}{x}\right) &= 2 \\ \frac{2x+1}{x} &= 25 \\ 2x+1 &= 25x \\ 23x &= 1 \Rightarrow \underline{x = \frac{1}{23}} \end{aligned}$$

Q 8

Solve $\ln(6-x) = 2\ln x$.

Sydney
Girls

- (A) $x = -3$ or $x = 2$ (B) $x = 6$ (C) $x = 2$ (D) $x = 2$ or $x = 6$

Q 9

Solve $\log_e(2x+2) + \log_e x - \log_e 12 = 0$

Sydney
Girls

(3)

$$\begin{aligned} \ln(2x+2)x - \log_e 12 &= 0 \\ \ln(2x^2+2x) &= \log_e 12 \\ 2x^2+2x-12 &= 0 \\ x^2+x-6 &= 0 \\ (x+3)(x-2) &= 0 \\ x &= -3 \text{ or } x = 2 \\ x &\neq -3 \\ \therefore \underline{x = 2} \end{aligned}$$



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PART C: SOLVING EQUATIONS

2 unit

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Q 10

Solve the following equation: $\log_2 x + \log_2(x+7) = 3$, for $x > 0$.

Sydney
Girls

$$\log_2 x + \log_2(x+7) = 3$$

$$\therefore \log_2 x(x+7) = 3$$

$$\frac{3}{2} = x(x+7)$$

$$8 = x^2 + 7x$$

$$x^2 + 7x - 8 = 0$$

$$(x+8)(x-1) = 0$$

$$x = -8 \quad \text{or} \quad x = 1$$

We reject $x = -8$ as $x > 0$

Q 11

Solve $\log_4 32 = x$.

2

Sydney
Girls

$$\log_4 32 = x$$

$$4^x = 32$$

$$2^{2x} = 2^5$$

$$\therefore 2x = 5$$

$$x = \frac{5}{2}$$

Q 12

Solve $(\log_{10} x^3)(\log_{10} x) + \log_{10} x^4 - 7 = 0$

3

Sydney
Girls

$$(\log_{10} x^3)(\log_{10} x) + \log_{10} x^4 - 7 = 0$$

$$3\log_{10} x + 4\log_{10} x - 7 = 0 \quad \checkmark$$

$$\text{Let } u = \log_{10} x$$

$$\therefore 3u^2 + 4u - 7 = 0 \quad \checkmark$$

$$u = 1 \quad \therefore \log_{10} x = 1$$

$$u = -\frac{7}{3} \quad \therefore \log_{10} x = -\frac{7}{3}$$

$$\therefore \begin{cases} x = 10 \\ x = 10^{-7/3} \end{cases} \quad \checkmark$$

Q 13

Solve $(\log_3 x)^2 - \log_3 x^4 - 12 = 0$, for $x > 0$.

[3]

Sydney
Girls

$$(\log_3 x)^2 - 4\log_3 x - 12 = 0 \quad \checkmark$$

$$\text{Let } u = \log_3 x$$

$$u^2 - 4u - 12 = 0$$

$$\begin{cases} u = 6 \\ u = -2 \end{cases} \quad \therefore \begin{cases} \log_3 x = 6 \\ \log_3 x = -2 \end{cases} \quad \therefore \begin{cases} x = 729 \quad \checkmark \\ x = \frac{1}{9} \quad \checkmark \end{cases}$$



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PART C: SOLVING EQUATIONS

Q 14

Solve the following:
 $\log_2 x + \log_2 (x+7) = 3$ for $x > 0$

3

Sydney
Tech

$$\log_2 x + \log_2 (x+7) = 3$$

$$\underline{x > 0}$$

$$x^2 + 7x - 8 = 0$$

$$\log_2 [x(x+7)] = 3$$

$$(x+8)(x-1) = 0$$

$$[x(x+7)] = 2^3$$

$$x = -8$$

$$\underline{x = 1}$$

$$2(x+7) = 8$$

↑
No Solution

↑
only solution

Q 15

Solve $2 \log_5 x - \log_5 (x+2) = \frac{2}{3} \log_5 125$

3

Sydney
Tech

$$2 \log_5 x - \log_5 (x+2) = \frac{2}{3} \log_5 125$$

$$\log_5 \left(\frac{x^2}{x+2} \right) = \log_5 (5^3)^{2/3}$$

$$\log_5 \left(\frac{x^2}{x+2} \right) = \log_5 5^2$$

$$\frac{x^2}{x+2} = 25$$

$$x^2 - 25x - 50 = 0$$

$$x = \frac{25 \pm \sqrt{825}}{2}$$

$$\text{As } x > 0$$

$$\underline{x = \frac{25 + \sqrt{825}}{2} \text{ only}}$$

Q 16

Given that $\log_{16} 2 = \log_x 3$, find x .

3

Enterprise

$$\log_{16} 2 = \log_x 3$$

$$\frac{\log_e 2}{\log_e 16} = \frac{\log_e 3}{\log_e x}$$

$$\frac{\log_e 2}{\log_e 2^4} = \frac{\log_e 3}{\log_e x}$$

$$\frac{\log_e 2}{4 \log_e 2} = \frac{\log_e 3}{\log_e x}$$

$$\frac{1}{4} = \frac{\log_e 3}{\log_e x}$$

$$\log_e x = 4 \log_e 3$$

$$\log_e x = \log_e 81$$

$$\underline{x = 81}$$

Q 17

On the diagram, AB is a vertical interval between $y = e^x(1+e)$ and $y = e^{2x}$. The length of $AB = e$ units.

(i) Show that $e^{2x} - e^x(1+e) + e = 0$.

(ii) What are the possible values of x that will keep the length of AB , e units?

1

3

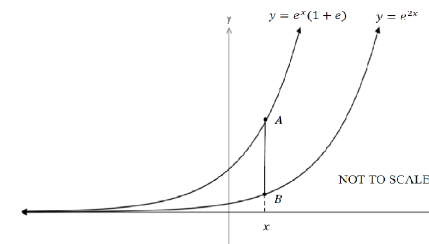
Santa
Sabina

$$\begin{aligned} \text{(2d(i)) distance } AB &= e \\ e^x(1+e) - e^{2x} &= e \\ 0 &= e^{2x} - e^x(1+e) + e \end{aligned}$$

$$\begin{aligned} \text{(iii)} \quad e^{2x} - e^x(1+e) + e &= 0 \\ \text{let } m &= e^x \end{aligned}$$

$$\begin{aligned} m^2 - m(1+e) + e &= 0 \\ (m-e)(m-1) &= 0 \\ \therefore m &= e \quad m = 1 \end{aligned}$$

$$\begin{aligned} \text{but } m &= e^x \\ \therefore e^x &= e \quad e^x = 1 \\ \therefore \underline{x = 1} \quad \text{OR} \quad \underline{x = 0} \end{aligned}$$





LOGS and EXPO selective

PART C: SOLVING EQUATIONS

2 unit

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Q 18

Find the solutions to the equation :

$$e^x \log_x 2 - 3e^x = 4 \log_x 2 - 12$$

3

Sydney

Girls

Express the solutions in exact form.

$$\begin{aligned} e^x \log_x 2 - 4 \log_x 2 &= 3e^x - 12 \\ \log_x 2 (e^x - 4) &= 3(e^x - 4) \\ \log_x 2 (e^x - 4) - 3(e^x - 4) &= 0 \\ (e^x - 4)(\log_x 2 - 3) &= 0 \\ e^x - 4 = 0 &\rightarrow e^x = 4 \\ x \ln e &= \ln 4 \\ \boxed{x = \ln 4} \\ \log_x 2 = 3 &\therefore x^3 = 2 \therefore x = \sqrt[3]{2} \end{aligned}$$

Q 19

Solve for x: $4^x - 5 \times 2^x + 4 = 0$

Barker

$$\begin{aligned} 2^{2x} - 5 \times 2^x + 4 &= 0 \\ m^2 - 5m + 4 &= 0 \\ (m-1)(m-4) &= 0 \quad m=1, 4 \\ 2^x &= 2^0, 2^2 \quad x=0, 2 \end{aligned}$$

Q 20

Solve $\ln 4 = 2 \ln x$.

Barker

$$\begin{aligned} \ln 4 &= 2 \ln x \\ \ln 4 &= \ln x^2 \\ 4 &= x^2 \\ x &= \pm 2 \\ \text{But } x > 0 &\therefore x=2 \\ &\text{is the only solution.} \end{aligned}$$



LOGS and EXPO selective

Part D: graphing

2 unit

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Q 1

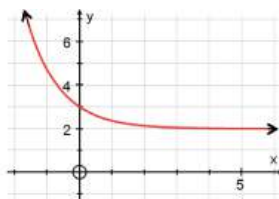
What is the equation of the graph shown below?

- (A) $y = -e^x + 2$ (C) $y = e^{-x} - 2$

Santa

Sabina

- (B) $y = e^{-x} + 2$ (D) $y = 2 - e^x$



Q 2

Which statement is **incorrect** for the graph of $y = -e^x$.

- (A) As $x \rightarrow \infty$ $y \rightarrow -\infty$ (B) As $x \rightarrow -\infty$ $y \rightarrow 0$

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- (C) When $x = 0$ $y = -1$ (D) As $x \rightarrow \infty$ $y \rightarrow \infty$

Q 3

Let $f(x) = 1 + e^x$. What is the range of $f(x)$?

- (A) $y > 0$ (B) $y \geq 0$ (C) $y > 1$ (D) $y \geq 1$

St

Catherines

Q 4

Which of the following is not true of the function $f(x) = e^x + 3$?

- (A) The first derivative is e^x .
(B) The function is always increasing.
(C) The function has its y -intercept at $(0, 1)$.
(D) A primitive of the function is $e^x + 3x$.

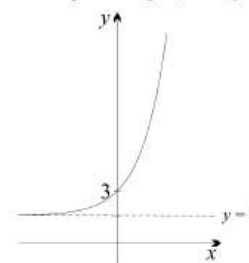
Sydney

Grammar

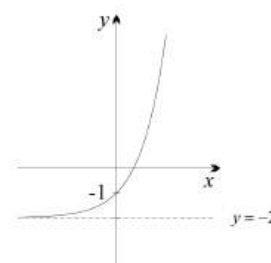
Q 5

Which of the following is the graph of $y = e^{-x} + 2$?

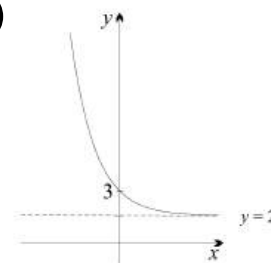
(A)



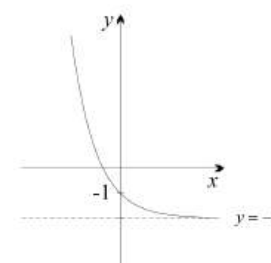
(B)



(C)



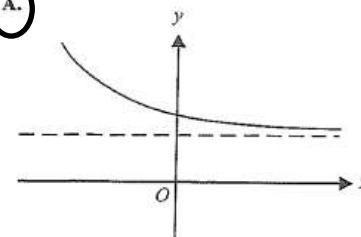
(D)



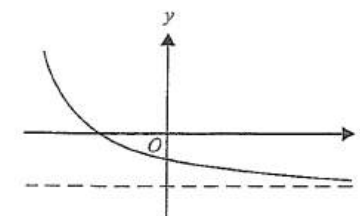
Q 6

If k is a negative real number and P is a positive real number, which one of the following is most likely to be the graph of the function with equation $y = e^{kx} + P$?

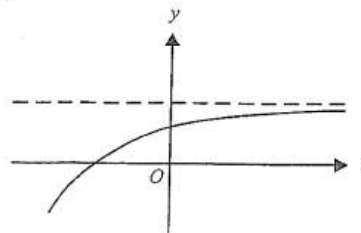
A.



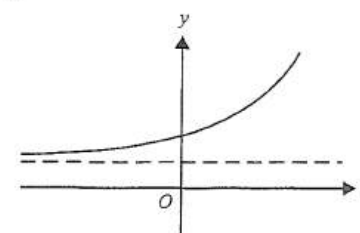
B.



C.



D.





LOGS and EXPO selective

PART D: graphing

2 unit

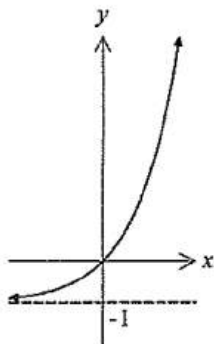
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Q 7

Sydney
Tech

What is a possible equation for the following graph?



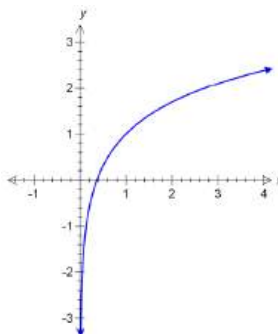
- (A) $y = e^{x-1}$
 (B) $y = e^x + 1$
 (C) $y = e^x - 1$
 (D) $y = e^{x+1}$

Q 8

St Ignatius

What is the equation of the graph below?

- (A) $y = \ln x$ (B) $y = 1 + \ln x$ (C) $y = \ln(x+1)$ (D) $y = \frac{1}{e^x}$

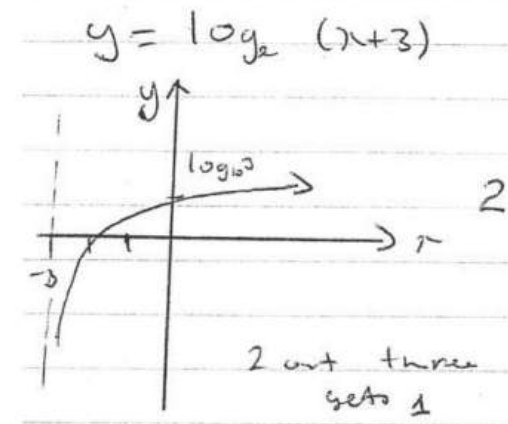


Q 9

Sydney
Girls

Sketch $y = \log_{10}(x+3)$, showing the x and the y intercepts.

[2]

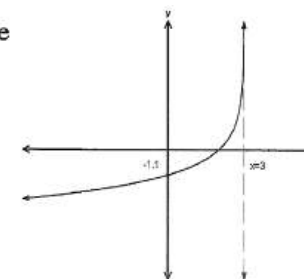


Q 10

Sydney
Tech

The equation of the graph shown could be

- (A) $y = \log_e(3-x)$
 (B) $y = -\log_e(3-x)$
 (C) $y = e^{(x-2)}$
 (D) $y = -\log_e(3x-1)$





LOGS and EXPO selective

Part D: graphing

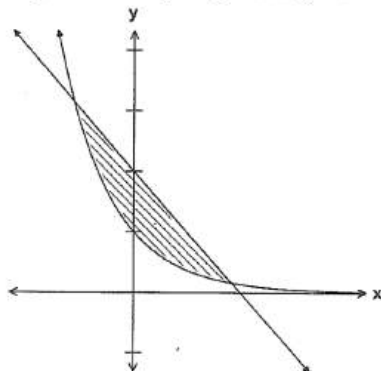
2 unit

TEACHER'S COPY

Q11

The diagram shows the region enclosed by $x + y = 2$ and $y = e^{-x}$

Sydney
Tech



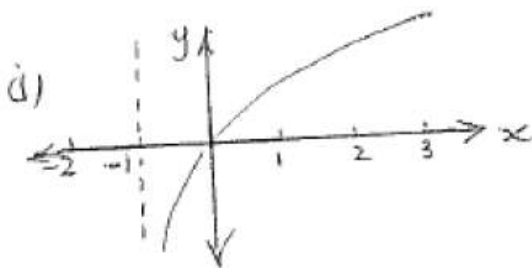
Which of the following pair of inequalities describes the shaded region in the diagram?

- (A) $x + y \leq 2$ and $y \leq e^{-x}$ (C) $x + y \geq 2$ and $y \leq e^{-x}$
 (B) $x + y \leq 2$ and $y \geq e^{-x}$ (D) $x + y \geq 2$ and $y \geq e^{-x}$

Q12

Sketch the curve $y = \ln(x+1)$ showing essential features.

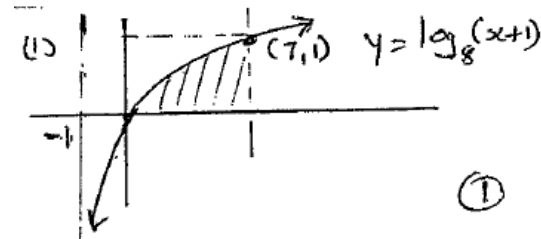
Barker



Q13

Sketch the graph $y = \log_8(x+1)$

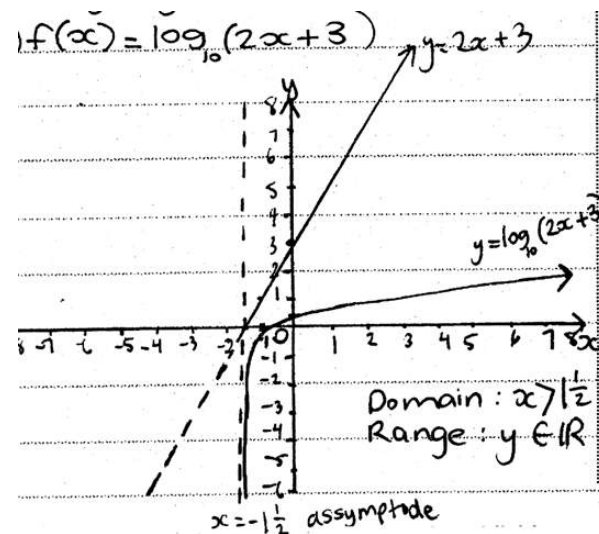
Baulkham
Hills



Q14

Sketch the graphs of $f(x) = \log_{10}(2x+3)$ stating its largest possible domain and range.

Baulkham
Hills



LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 1

If $y = \frac{e^x}{2}$ find the gradient of the tangent at the point $(0, 1)$?

Santa
Sabina

$$\begin{aligned} \text{a) } y' &= \frac{e^x}{2} \\ \text{At } (0, 1) \text{ } m &= \frac{e^0}{2} \\ &= \frac{1}{2} \end{aligned}$$

Q 2

Differentiate the following functions. Leave answers in simplified form

(i) $y = e^{\tan x}$

Santa
Sabina

(ii) $y = \frac{e^{2x-3}}{x^2}$

$$\begin{aligned} \text{b) (i) } y' &= \sec^2 x \cdot e^{\tan x} \\ \text{(ii) } y &= \frac{e^{2x-3}}{x^2} \\ y' &= \frac{x^2 \cdot 2e^{2x-3} - 2x \cdot e^{2x-3}}{x^4} \\ &= \frac{2xe^{2x-3}(x-1)}{x^4} \\ &= \frac{2e^{2x-3}(x-1)}{x^3} \end{aligned}$$

Q 3

What is $\frac{d}{dx}(2e^{2\pi x})$ equivalent to?

- (A) $2e^{2\pi x}$ (B) $2\pi e^{2\pi x}$ (C) $4e^{2\pi x}$ (D) $4\pi e^{2\pi x}$

Santa
Sabina

[2]

Q 4

What is the first derivative of the function $x^3 e^x$ with respect to x ?

Santa
Sabina

- (A) $3x^2 e^x$ (C) $3x^2 e^x - e^x x^3$
(B) $e^x x^2 (3 + x)$ (D) $\frac{3x^2 - x^3}{e^x}$

Q 5

Find the equation of the normal to the curve $y = e^{2x}$ at the point $(\frac{1}{2}, e)$.

Santa
Sabina

$$\begin{aligned} y &= e^{2x} \\ y' &= 2e^{2x} \quad \text{①} \\ \text{m of tangent at } x = \frac{1}{2} &= 2e^{2 \cdot \frac{1}{2}} \\ &= 2e \quad \text{②} \\ \therefore \text{m of normal} &= -\frac{1}{2e} \quad \text{③} \\ \text{eqn of normal } y - e &= -\frac{1}{2e} \left(x - \frac{1}{2}\right) \quad \text{④} \\ 2ey - 2e^2 &= -x + \frac{1}{2} \\ x + 2ey - 2e^2 - \frac{1}{2} &= 0 \end{aligned}$$

Q 6

What is the derivative of $y = e^{x^2}$?

Santa
Sabina

- (A) $y' = 2x e^{x^2}$ (B) $y' = 2 e^{x^2}$
(C) $y' = 2 x e^x$ (D) $y' = 2 e^x$

LOGS and EXPO selective

Part e: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 7

Differentiate the following functions leaving answers in simplified form.

Santa
Sabina

(i) $y = x e^{2x+3}$

(ii) $y = \frac{e^{3x-1}}{x^3}$

1) $y = x e^{2x+3}$
 $u = x \quad u' = 1$
 $v = e^{2x+3} \quad v' = 2e^{2x+3}$
 $y' = 2x e^{2x+3} + e^{2x+3}$
 $\therefore y' = e^{2x+3} (2x+1)$

2) $y = \frac{e^{3x-1}}{x^3}$
 $u = e^{3x-1} \quad u' = 3e^{3x-1}$
 $v = x^3 \quad v' = 3x^2$
 $y' = \frac{3x^3 e^{3x-1} - 3x^2 e^{3x-1}}{x^6}$
 $= \frac{3x^2 e^{3x-1} (x-1)}{x^6}$
 $\therefore y' = \frac{3e^{3x-1} (x-1)}{x^4}$

Q 8

Find the equation of the normal to the curve $y = 2e^{3x^2+1}$ at $x = 1$.

Santa
Sabina

d) $y = 2e^{3x^2+1}$
 $y' = 12x e^{3x^2+1}$
 at $x=1$
 $y' = 12e^4$
 $\therefore m_{\text{normal}} = -\frac{1}{12e^4}$

at $x=1 \quad y = 2e^4$
 $y - 2e^4 = -\frac{1}{12e^4} (x-1)$
 $12e^4 - 24e^8 = 1 - x$
 $x + 12e^4 - 24e^8 - 1 = 0$

Q 9

What is the derivative of $y = \frac{e^{2x} + 4}{x^3}$?

Santa
Sabina

(A) $\frac{e^{2x}}{3x^2}$

(C) $\frac{2xe^{2x} - 3e^{2x} - 12}{x}$

(B) $\frac{2xe^{2x} - 3e^{2x} - 12}{x^4}$

(D) $\frac{2x^{2x} - 3e^{2x} + 12}{x^4}$

Q 10

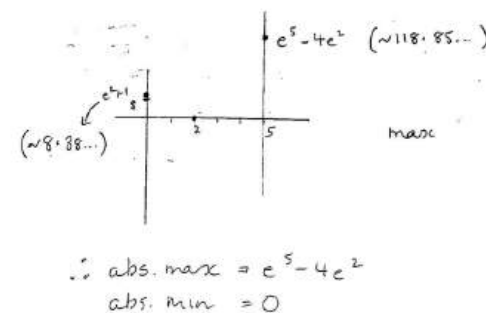
Santa
Sabina

Determine the absolute maximum and minimum values of y for the curve:

$y = e^2 + e^x - x e^2$ in the domain $0 \leq x \leq 5$.

(Give your answers as exact values.)

(c) $y = e^2 + e^x - x e^2$
 $y' = e^x - e^2$
 Set $y' = 0$ for stat pts
 $e^x - e^2 = 0$
 $x = 2$ (only 1 stat p)
 $y'' = e^x$
 when $x=2, y'' = e^2 > 0 \therefore$ min
 when $x=2, y = e^2 + e^2 - 2e^2 = 0$
 when $x=0, y = e^2 + e^0 - 0 = e^2 + 1$
 when $x=5, y = e^2 + e^5 - 5e^2 = e^5 - 4e^2$



LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 11

Differentiate $(2e^x - 3)^6$

St George
Girls

$$\begin{aligned}\frac{d}{dx}(2e^x - 3)^6 \\&= 6(2e^x - 3)^5 \cdot 2e^x \\&= 12e^x(2e^x - 3)^5\end{aligned}$$

Q 12

Differentiate:

St George
Girls

(i) $y = x^3 e^{3x}$ (ii) $y = \frac{e^x}{(x+3)^2}$

(i) $y = \sin^2 4x$
 $y = [\sin(4x)]^2$
 $\frac{dy}{dx} = 2 \cdot \sin(4x) \times \cos(4x) \times 4$
 $= 8 \sin(4x) \cos(4x)$

(ii) $y = x^3 e^{3x}$ $u = x^3$ $v = e^{3x}$
 $\frac{dy}{dx} = v \frac{du}{dx} + u \frac{dv}{dx}$ $\frac{du}{dx} = 3x^2$ $\frac{dv}{dx} = 3e^{3x}$
 $= 3x^2 e^{3x} + 3x^3 e^{3x}$
 $= 3x^2 e^{3x} (1 + x)$

Q 13

Differentiate $(e^x - 3)^4$

St George
Girls

$$\begin{aligned}y &= (e^x - 3)^4 \\ \frac{dy}{dx} &= 4(e^x - 3)^3 \times e^x \\ &= 4e^x (e^x - 3)^3\end{aligned}$$

Q 14

The derivative of e^{x^3} is:

St George
Girls

(A) $x^2 e^{x^3} (3 + x)$ (B) e^{x^3} (C) $3x^2 e^{x^3}$ (D) $\frac{e^{x^3}}{3x^2}$

Q 15

Differentiate $\sqrt{e^{2x} + 4}$

St George
Girls

$$\begin{aligned}y &= \sqrt{e^{2x} + 4} \\ y &= (e^{2x} + 4)^{\frac{1}{2}} \\ y' &= \frac{1}{2}(e^{2x} + 4)^{-\frac{1}{2}} \times 2e^{2x} \\ &= e^{2x} (e^{2x} + 4)^{-\frac{1}{2}}\end{aligned}$$

Q 16

Differentiate $(6e^{2x} + 2)^5$

St Ignatius

$$\begin{aligned}&5(6e^{2x} + 2)^4 \times 12e^{2x} \\ &= 60e^{2x} (6e^{2x} + 2)^4\end{aligned}$$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 17

Sydney
Boys

For the curve $y = xe^{-x}$,

- Prove that $\frac{dy}{dx} = e^{-x}(x-1)$.
- Find any stationary points and determine their nature.
- Prove that $\frac{d^2y}{dx^2} = e^{-x}(x-2)$.
- Show that there is a point of inflexion on this curve and find the co-ordinates of this point.
- Sketch the curve, showing the co-ordinates of the point of inflexion and any stationary points.

(i) $\frac{dy}{dx} = 1 \times e^{-x} + x \times (-1) \times e^{-x}$ (using the product rule),
 $= e^{-x}(1-x)$,
 $= -e^{-x}(x-1)$.

(ii) $\frac{dy}{dx} = 0$ when $x = 1$.

$\therefore$ Maximum at $(1, 1/e)$.

	0	1	2
$\frac{dy}{dx}$	1	0	-0.135
	$\nearrow$	$\rightarrow$	$\searrow$

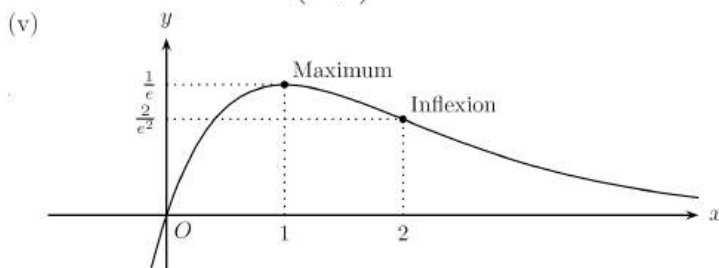
(iii) $\frac{d^2y}{dx^2} = -(-1)(e^{-x})(x-1) + 1 \times (-e^{-x})$ (product rule),
 $= e^{-x}(x-1) - e^{-x}$,
 $= e^{-x}(x-2)$.

(iv) $\frac{d^2y}{dx^2} = 0$ when $x = 2$.

Change of concavity.

$\therefore$ Inflexion at $(2, \frac{2}{e^2})$.

	1	2	3
$\frac{d^2y}{dx^2}$	-0.37	0	0.05
	$\downarrow$		$\uparrow$



Q 18

Sydney
Boys

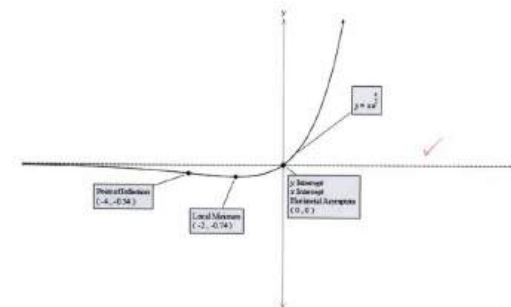
Sketch the curve $y = xe^{\frac{x}{2}}$ showing any asymptotes and stationary points. (3)
 It is NOT necessary to find any points of inflexion.

Intercepts $x=0, y=0$
 $y = xe^{\frac{x}{2}}$
 $y' = e^{\frac{x}{2}}(\frac{x}{2} + 1)$
 $y' = 0, x = -2, y = -\frac{2}{e}$

x	-3	-2	0
y'	-	0	+

 $\therefore$ min at $(-2, -\frac{2}{e})$

$x \rightarrow \infty, y \rightarrow \infty$
 $x \rightarrow -\infty, y \rightarrow 0 \therefore$ hor. asy at $y=0$
 x -intercept $= 0, y$ -intercept $= 0$



Q 19

Sydney
Boys

If $y = \frac{3}{x^2} - 2e^{-x+4}$, then $\frac{dy}{dx}$ is equal to

- (A) $\frac{-6}{x} + 2e^{-x+4}$ (B) $\frac{-6}{x^3} - 2e^{-x+4}$ (C) $\frac{-6}{x} - 2e^{-x+4}$ (D) $\frac{-6}{x^3} + 2e^{-x+4}$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 20

Sydney
Boys

What is the derivative of $(e^{3x} + 1)^{-2}$?

- A. $-2e^{3x}(e^{3x} + 1)^{-3}$ C. $-6e^{3x}(e^{3x} + 1)^{-3}$
B. $-2e^{3x}(e^{3x} + 1)^{-1}$ D. $-6e^{3x}(e^{3x} + 1)^{-1}$

Q 21

Sydney
Boys

Suppose that $y = e^{kx}$

- (i) Find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$
(ii) Find the value of k such that $y = 2\frac{dy}{dx} - \frac{d^2y}{dx^2}$

$$1) y = e^{kx}$$

$$\frac{dy}{dx} = ke^{kx} \quad (1)$$

$$\frac{d^2y}{dx^2} = k^2e^{kx} \quad (1)$$

$$(ii) y = 2\frac{dy}{dx} - \frac{d^2y}{dx^2}$$

$$e^{kx} = 2ke^{kx} - k^2e^{kx}$$

$$k^2e^{kx} - 2ke^{kx} + e^{kx} = 0$$

$$e^{kx}(k^2 - 2k + 1) = 0$$

$$e^{kx} \neq 0$$

$$\therefore k^2 - 2k + 1 = 0$$

$$(k-1)^2 = 0$$

$$k = 1 \quad (1)$$

Q 22

Sydney
Girls

Find the equation of the tangent to the curve $y = 2xe^x$ at the point $(1, e)$.

$$y = 2xe^x$$

$$\text{let } u = 2x, v = e^x$$

$$\frac{du}{dx} = 2 \quad \frac{dv}{dx} = e^x$$

$$\therefore \frac{dy}{dx} = v\frac{du}{dx} + u\frac{dv}{dx}$$

$$= 2e^x + 2xe^x$$

$$\frac{dy}{dx} = 2e^x(1+x)$$

$$\text{at } x = 1$$

$$\frac{dy}{dx} = 2e(2)$$

$$m = 4e$$

$$\therefore \text{Eqn of tangent}$$

$$y - y_1 = m(x - x_1)$$

$$y - e = 4e(x - 1)$$

$$y - e = 4ex - 4e$$

$$y = 4ex - 3e$$

$$\text{or}$$

$$4ex - y - 3e = 0$$

$$(2)$$

Q 23

Sydney
Girls

Differentiate $y = (e^x + 3)^4$.

$$y = (e^x + 3)^4$$

$$y' = 4(e^x + 3)^3(e^x)$$

$$= 4e^x(e^x + 3)^3$$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 24

What is the derivative of $\frac{e^x}{x^2}$?

Sydney
Girls

(A) $\frac{e^x}{2x}$

(B) $\frac{3e^x}{x^3}$

(C) $\frac{e^x(x-2)}{x^3}$

(D) $\frac{e^x(x^2-2x)}{x^2}$

Q 25

Use the quotient rule to differentiate $y = \frac{2e^{2x+3}}{x+3}$. In your answer simplify the numerator as far as possible.

Sydney
Grammar

$$\begin{aligned} y' &= \frac{(x+3)4e^{2x+3} - 2e^{2x+3}}{(x+3)^2} \checkmark \\ &= \frac{e^{2x+3}(4(x+3) - 2)}{(x+3)^2} \\ &= \frac{e^{2x+3}(4x+10)}{(x+3)^2} \checkmark \end{aligned}$$

Q 26

Find the first and second derivatives of e^{x^2} .

3

Sydney
Grammar

$$\begin{aligned} y &= e^{x^2} \\ \frac{dy}{dx} &= 2xe^{x^2} \checkmark \\ \frac{d^2y}{dx^2} &= 2x \cdot 2xe^{x^2} + 2e^{x^2} \\ &= 4x^2e^{x^2} + 2e^{x^2} \\ &= 2e^{x^2}(2x^2+1) \checkmark \end{aligned}$$

Q 27

Find the equation of the normal to the curve $y = x + e^x$ at the point where the curve cuts the y-axis.

3

Sydney
Grammar

$$\begin{aligned} y &= x + e^x \\ \frac{dy}{dx} &= 1 + e^x \checkmark \\ \text{At } x=0 \\ \frac{dy}{dx} &= 1+1 \\ &= 2 \end{aligned}$$

So gradient of tangent is 2.
and $\therefore$ gradient of normal is $-\frac{1}{2}$ $\checkmark$
When $x=0$
 $y = 0 + e^0$
 $= 1$

$$\begin{aligned} y - y_1 &= m(x - x_1) \\ y - 1 &= -\frac{1}{2}(x - 0) \\ y - 1 &= -\frac{1}{2}x \\ 2y - 2 &= -x \\ \boxed{x + 2y - 2} &= 0 \checkmark \end{aligned}$$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 28

Sydney
Grammar

Find the equation of the tangent to the curve $y = e^{3x}$ at the point where $x = 0$.

3

$$y = e^{3x} \rightarrow y' = 3e^{3x}$$

$$\text{At } x=0, y'(0) = 3 \quad \checkmark$$

$$\text{Also, } y(0) = e^0 = 1 \quad \checkmark$$

$$\therefore \text{line is: } y - 1 = 3(x - 0)$$

$$y = 3x + 1 \quad \checkmark$$

$$\text{OR } 3x - y + 1 = 0 \quad \checkmark$$

Q 29

Sydney
Grammar

Using the quotient rule, or otherwise, find the derivative of $y = \frac{2x^2}{e^{3x}}$. Express your answer in simplest form.

$$y = \frac{2x^2}{e^{3x}}$$

$$\frac{dy}{dx} = \frac{e^{3x} \cdot 4x - 2x^2 \cdot 3e^{3x}}{(e^{3x})^2} \quad \checkmark$$

$$= \frac{4x - 6x^2}{e^{3x}} \quad \checkmark$$

$$= \frac{2x(2-3x)}{e^{3x}}$$

Q 30

Sydney
Grammar

Consider the exponential function $y = e^{-3x}$.

- (i) Find y' and y'' .
(ii) Show that $y = e^{-3x}$ satisfies the equation $3y = 5y' + 2y''$.

2

1

i) $y = e^{-3x}$
 $y' = -3e^{-3x} \quad \checkmark$
 $y'' = 9e^{-3x} \quad \checkmark$

ii) Show $3y = 5y' + 2y''$
 $\text{LHS} = 3 \times e^{-3x}$
 $= 3e^{-3x}$
 $\text{RHS} = 5 \times (-3e^{-3x}) + 2 \times 9e^{-3x}$
 $= -15e^{-3x} + 18e^{-3x}$
 $= 3e^{-3x}$
 $\therefore \text{LHS} = \text{RHS}$
 $\therefore 3y = 5y' + 2y''$ as req'd. $\checkmark$

Q 31

Sydney
Grammar

Find the equation of the tangent to the curve $y = e^{3x} + 1$ at the point $(0, 2)$.

2

$$\frac{dy}{dx} = 3e^{3x}$$

$$\text{when } x=0, \frac{dy}{dx} = 3$$

$$y - 2 = 3(x - 0)$$

$$\therefore y = 3x + 2$$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 32

Sydney
Grammar

Consider the curve $y = x^3 e^x$.

- Show that $\frac{dy}{dx} = 3e^x x^2 + x^3 e^x$.
- Find all stationary points on the curve. Give your answers in exact form.
- Using a table of gradients, or otherwise, determine the nature of each stationary point.
- What value does y approach as x approaches $-\infty$?
- Sketch the curve, clearly indicating the stationary points.

b) i) $y = x^3 e^x$

$\frac{dy}{dx} = vu' + uv'$

$u = x^3 \quad v = e^x$
 $u' = 3x^2 \quad v' = e^x$

✓ show.

$\frac{dy}{dx} = 3x^2 e^x + x^3 e^x$ as required.

ii) stationary points $\frac{dy}{dx} = 0$

$3x^2 e^x + x^3 e^x = 0$

$x^2 e^x (3 + x) = 0$

$x^2 (3 + x) = 0$ (since $e^x \neq 0$)

$x = 0 \quad x = -3$

at $x = 0 \quad y = 0$ at $x = -3 \quad y = -\frac{27}{e^3}$

penalise not exact

Stationary points: $(0, 0)$ $(-3, -\frac{27}{e^3})$

1

2

2

1

1

iii)

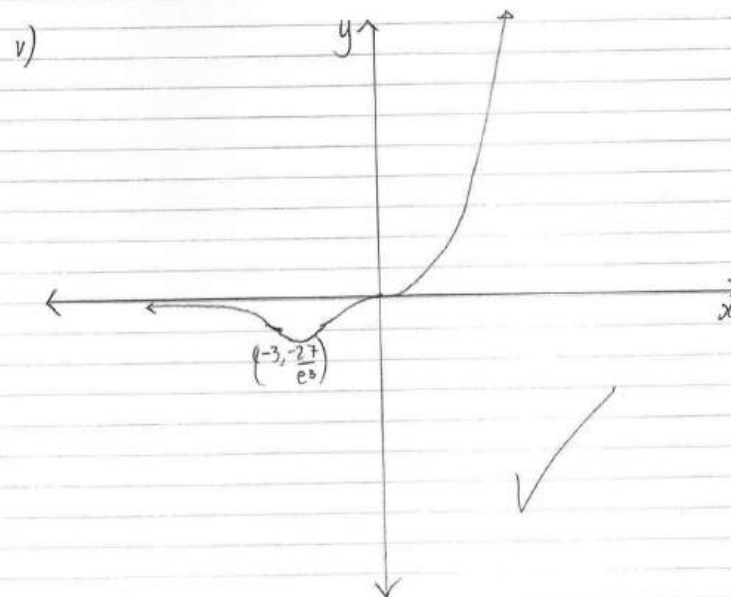
x	-4	-3	-1	0	1
$\frac{dy}{dx}$	-0.29	0	0.735...	0	10.873

✓ needs numeric values

$(-3, -\frac{27}{e^3})$ is a minimum turning point ✓/conclude

$(0, 0)$ is a horizontal inflexion point.
penalise inflex point

iv) as $x \rightarrow -\infty \quad y \rightarrow 0$ ✓



LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 33

Which of the following does $\frac{d}{dx}(e^3)$ equal?

Sydney
Grammar

- (A) $3e^2$ (B) e^3 (C) 0 (D) $\frac{1}{4}e^4$

Q 34

Differentiate $y = \frac{e^{5x} + 1}{e^x}$.

2

Sydney
Grammar

$$y = \frac{e^{5x} + 1}{e^x}$$

$$= e^{4x} + e^{-x}$$

$$\text{so } \frac{dy}{dx} = 4e^{4x} - e^{-x}$$

Q 35

Which of the following is the derivative of $y = \frac{e^{7x}}{e^{3x}}$?

Sydney
Grammar

(A) $4e^{4x}$

(B) e^{4x}

(C) $\frac{7e^{3x}e^{7x} + 3e^{3x}e^{7x}}{e^{9x}}$

(D) $\frac{3e^{3x}e^{7x} - 7e^{7x}e^{3x}}{e^{9x}}$

Q 36

If $g(t) = e^{-t} - 1$ then $g'(0)$ equals

Sydney
Tech

- A. $-e$ B. -2 (C) -1 D. 0

Q 37

Differentiate $(x+2)e^{2x}$ 2

Sydney
Tech

$$e^{2x} + 2(x+2)e^{2x}$$

$$= e^{2x}(2x+5)$$

Q 38

Find the equation of the tangent to $y = e^{4x} + x$ at the point where $x = 0$. 3

Sydney
Tech

$$y = e^{4x} + x$$

$$y' = 4e^{4x} + 1$$

when $x=0$

$$y' = 4e^0 + 1$$

$$= 5$$

$$y = e^0 + 0$$

$$= 1$$

$\therefore m=5$ passing thru' (0,1)

$$y - y_1 = m(x - x_1)$$

$$y - 1 = 5(x - 0)$$

$$5x - y + 1 = 0$$

LOGS and EXPO selective

Part E: Differentiation of exponentials

2 unit

TEACHER'S COPY

Q 39

For what values of k does $y = 3e^{kx}$ satisfy the equation $\frac{d^2y}{dx^2} - 9y = 0$?

Sydney

Tech

$$\begin{aligned} y &= 3e^{kx} \\ y' &= 3ke^{kx} \\ y'' &= 3k^2e^{kx} \\ \therefore y'' - 9y &= 0 \\ 3k^2e^{kx} - 27e^{kx} &= 0 \\ 3e^{kx}(k^2 - 9) &= 0 \\ \therefore k &= \pm 3 \end{aligned}$$

Q 40

What is the derivative of $e^x(x^2 + 2x)$?

Sydney

Tech

- (A) $(2x + 2)$ (C) $e^x(x^2 - 2)$
(B) $e^x(2x + 2)$ (D) $e^x(x^2 + 4x + 2)$

Q 41

Find the equation of the tangent to the curve $y = 3e^x - 1$ at the point where $x = 1$

Sydney

Tech

$$\begin{aligned} y &= 3e^x - 1 \\ \frac{dy}{dx} &= 3e^x \\ \text{When } x=1 \quad y &= 3e - 1 \\ m &= 3e \\ \text{Equation of tangent} \\ y - (3e - 1) &= 3e(x - 1) \\ y - 3e + 1 &= 3ex - 3e \\ y + 1 &= 3ex \\ y &= 3ex - 1 \end{aligned}$$

Q 42

Given $y = e^{kx}$, find the value of k such that $y = 2\frac{dy}{dx} - \frac{d^2y}{dx^2}$

Sydney

Tech

$$\begin{aligned} y &= e^{kx} \quad y' = ke^{kx} \quad y'' = k^2e^{kx} \\ y &= 2y' - y'' \\ e^{kx} &= 2(ke^{kx}) - (k^2e^{kx}) \\ 1 &= 2k - k^2 \\ k^2 - 2k + 1 &= 0 \\ (k - 1)^2 &= 0 \\ \therefore k &= 1 \end{aligned}$$



LOGS and EXPO selective

PART F: general Application

2 unit

TEACHER'S COPY



Q 1

The equation of the tangent to the curve $f(x) = 2e^x(x^2+1)$ at the point $(0, 2)$

Barker

is given by $y = ax + b$

Evaluate a and b .

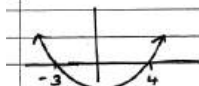
$$\begin{aligned} c) f(x) &= 2e^x(x^2+1) \\ f'(x) &= 2e^x(x^2+1) + 2e^x \times 2x \\ &= 2e^x(x^2+2x+1) \\ f'(0) &= 2(1) = 2 \\ x=0 \quad f(0) &= 2(1) = 2 \\ y-2 &= 2(x-0) \\ y &= 2x+2 \\ \therefore a &= 2, \quad b = 2 \end{aligned}$$

Q 2

Find all values of k for which $y = e^{kx}$ is a solution of $y'' - y' - 12y \geq 0$

Baulkham
Hills

$$\begin{aligned} y &= e^{kx} \\ y' &= ke^{kx} \quad (1) \\ y'' &= k^2 e^{kx} \quad (1) \\ [\div e^{kx} \text{ and factorise}] & \quad k^2 - k - 12 \geq 0 \\ & \quad e^{kx}(k^2 - k - 12) \geq 0 \\ & \quad (k-4)(k+3) \geq 0 \quad (1) \\ \text{Roots: } k &= 4, -3 \\ \text{Solu: } & \\ k &\geq 4, \quad k \leq -3 \quad (1) \end{aligned}$$

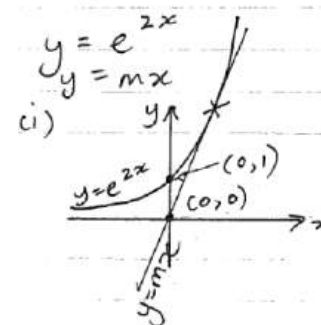


Q 3

The line $y = mx$ is a tangent to the curve $y = e^{2x}$.

Barker

- Draw a sketch of $y = mx$ and $y = e^{2x}$ on the same set of axes, showing this information.
- Find the value of m .



$$\begin{aligned} (ii) \quad e^{2x} &= mx \\ \ln(e^{2x}) &= \ln mx \\ 2x &= \ln mx \quad \dots (1) \\ y' &= 2e^{2x} \\ m &= 2e^{2x} \quad \dots (2) \\ 2x &= \ln 2xe^{2x} \quad \text{② into ①} \\ e^{2x} &= 2xe^{2x} \\ 1 &= 2x \\ x &= \frac{1}{2} \\ \therefore m &= 2e \\ * \text{ There are other } * \\ &\quad \text{methods} \end{aligned}$$



LOGS and EXPO selective

PART F: general APPLICATION

2 unit

TEACHER'S COPY



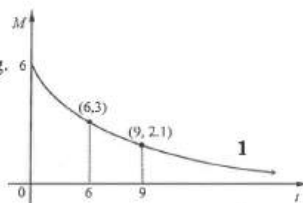
Q 4

CSSA

A drug is used to control a medical condition for a human body. The graph below shows the amount M (in milligrams) of the drug remaining in the body after t hours.

The graph above is modelled by the equation $M = Pe^{Kt}$.

Use the information given in the graph to answer the following.



- Find the value of P . 2
- Hence find the value of K , correct to 4 decimal places. 2
- When will one-fifth of the initial dose remain? Write your answer correct to the nearest hour. 2
- At what rate is the drug being removed from the human body when $t = 9$ hours? 2

From the graph, $P = 6$.

When $t = 6$, $M = 3$ and $P = 6$:

$$3 = 6e^{6k}$$

$$\frac{1}{2} = e^{6k}$$

$$\ln\left(\frac{1}{2}\right) = 6k$$

$$k \approx -0.1155 \text{ (4 decimal places)}$$

$$1.2 = 6e^{-0.1155 \times t}$$

$$\frac{1.2}{6} = e^{-0.1155 \times t}$$

$$\ln\left(\frac{1.2}{6}\right) = -0.1155 \times t$$

$$t \approx 13.934 \dots$$

So one fifth of the initial dose will remain after approximately 14 hours.

$$\frac{dM}{dt} = kPe^{kt}$$

$$= -0.1155 \times 6 \times e^{-0.1155 \times 9}$$

$$= -0.245 \text{ mg/hr}$$

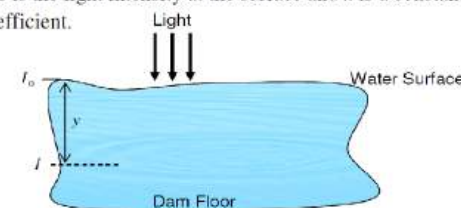
$\therefore$ Drug removed at rate of 0.245 mg/hr.

Q 5

CSSA

The light intensity, I units, passing through y metres of water is given by the equation: $I = I_0 e^{-ky}$ $y \geq 0$

where I_0 units is the light intensity at the surface and k is a constant called the absorption coefficient.



Above is the cross-section of Warragamba Dam and in the table below are the readings of two light intensity measurements from the dam.

y metres	I Units
2	1.2
8	0.9

- Using the table show that $k = \frac{1}{6} \log_e\left(\frac{4}{3}\right)$, and hence find I_0 . 3
- Sketch the graph $I = I_0 e^{-ky}$ indicating the vertical intercept. 2
- The fish that live in this dam require light in order to survive. One kind of fish requires light of an intensity that is no less than 35% of the intensity at the surface. Determine the maximum depth, correct to the nearest metre, at which this kind of fish can survive. 2

Sample Answer:

$$I = I_0 e^{-ky}$$

$$\therefore \text{from the table } 1.2 = I_0 e^{-2k} \dots \textcircled{1} \text{ and } 0.9 = I_0 e^{-8k} \dots \textcircled{2}$$

$$\therefore \textcircled{1} \div \textcircled{2} \quad \frac{1.2}{0.9} = \frac{e^{-2k}}{e^{-8k}} \quad \therefore \frac{4}{3} = e^{6k} \quad \log_e \frac{4}{3} = 6k$$

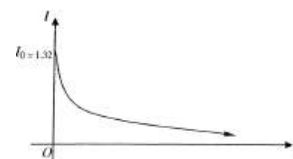
$$\therefore k = \frac{1}{6} \log_e\left(\frac{4}{3}\right) = 0.047947$$

$$1.2 = I_0 e^{-2 \times \frac{1}{6} \log_e\left(\frac{4}{3}\right)}$$

$$\therefore 1.2 = I_0 \times 0.90856$$

$$I_0 = 1.3207709 = 1.32$$

Sample Answer:



Sample Answer:

$$\text{If } I > 35\% I_0 \quad \therefore I_0 e^{-0.04795y} \geq 0.35 I_0$$

$$\therefore -0.04795y \geq \ln 0.35$$

$$\therefore y \leq \frac{\ln 0.35}{-0.04795} \quad \text{i.e. } y \leq 21.8$$

$\therefore$ The maximum depth at which this kind of fish can survive is 21 metres.